[nmap]

nmap -sn [IP] - 가장 기본 스캔

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -sn 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 19:56 KST
Nmap scan report for 172.16.29.11
Host is up (0.00070s latency).
MAC Address: 00:0C:29:ED:03:6B (VMware)
Nmap done: 1 IP address (1 host up) scanned in 0.15 seconds
```

호스트 살아있는지 확인
→ ping sweep(핑 스캔). OS는 못잡음. 살아있는지 여부만.

nmap -sV [IP] - 서비스 스캔(가장 많이 씀)

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -sV 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 19:57 KST
Nmap scan report for 172.16.29.11
Host is up (0.00016s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
21/tcp   open  ftp     vsftpd 3.0.5
22/tcp   open  ssh     OpenSSH 8.7 (protocol 2.0)
53/tcp   open  domain  ISC BIND 9.16.23 (RedHat Linux)
80/tcp   open  http    Apache httpd 2.4.62 ((Rocky Linux))
3306/tcp open  mysql   MariaDB 5.5.5-10.5.29
8000/tcp open  http    Tornado httpd 6.5.1
MAC Address: 00:0C:29:ED:03:6B (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.26 seconds
```

→ 열려있는 포트 + 서비스 버전
SSH인지 SFTP인지, Apache인지 Nginx인지, DB 버전까지 다 나옴.

nmap -O [IP] - 운영체제(OS) 탐지

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -O 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 19:58 KST
Nmap scan report for 172.16.29.11
Host is up (0.00056s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
3306/tcp open  mysql
8000/tcp open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.79 seconds
```

→ OS guess (정확도는 환경 따라 다름)
※ -sn 과는 별개. -sn에는 OS 정보 포함 X.

실무에서 '정확한 OS 확인'은 보통 Banner나 HTTP Header로 더 확인함.

nmap -p- [IP] - 포트 전체 스캔

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -p- 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:01 KST
Nmap scan report for 172.16.29.11
Host is up (0.00042s latency).
Not shown: 65529 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
3306/tcp open  mysql
8000/tcp open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 2.63 seconds
```

→ 0~65535 전체 스캔
웹서버 숨겨진 포트 찾을 때 필수.

nmap -sV -sC -O -p- [IP] - 강력 조합(실무에서 제일 자주 씀)

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -sV -sC -O -p- 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:03 KST
Nmap scan report for 172.16.29.11
Host is up (0.00080s latency).
Not shown: 65529 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
21/tcp   open  ftp     vsftpd 3.0.5
22/tcp   open  ssh     OpenSSH 8.7 (protocol 2.0)
| ssh-hostkey:
|   256 9b:5c:db:d0:44:c3:cb:3e:ad:f5:98:4b:ac:3a:e8:3c (ECDSA)
|_  256 a8:ed:13:f8:62:c5:82:c8:ac:40:ed:a3:5e:01:f9:a4 (ED25519)
53/tcp   open  domain  ISC BIND 9.16.23 (RedHat Linux)
| dns-nsid:
|_  bind.version: 9.16.23-RH
80/tcp   open  http    Apache httpd 2.4.62 ((Rocky Linux))
|_http-server-header: Apache/2.4.62 (Rocky Linux)
|_http-title: PHP 8.4.14 - phpinfo()
3306/tcp open  mysql   MariaDB 5.5.5-10.5.29
| mysql-info:
|   Protocol: 10
|   Version: 5.5.5-10.5.29-MariaDB
|   Thread ID: 1669
|   Capabilities flags: 63486
|   Some Capabilities: Support41Auth, SupportsCompression, ODBCClient, IgnoreSigpipes, Speaks41P
rotocolOld, ConnectWithDatabase, InteractiveClient, IgnoreSpaceBeforeParenthesis, LongColumnFlag
, DontAllowDatabaseTableColumn, Speaks41ProtocolNew, SupportsTransactions, SupportsLoadDataLocal
, FoundRows, SupportsAuthPlugins, SupportsMultipleStatments, SupportsMultipleResults
|   Status: Autocommit
|   Salt: 1}E!q;qFHyH:+fBVk9wl
|_  Auth Plugin Name: mysql_native_password
8000/tcp open  http    Tornado httpd 6.5.1
|_http-server-header: TornadoServer/6.5.1
| http-title: Jupyter Server
|_Requested resource was /login?next=%2Ftree%3F
MAC Address: 00:0C:29:ED:03:6B (VMware)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/subm
it/ .
Nmap done: 1 IP address (1 host up) scanned in 18.74 seconds
```

-sV : 버전
-sC : 기본 NSE 스크립트
-O : OS
-p- : 전체 포트
→ "이 서버가 뭐하는 애냐?" 한방에 끝.

nmap -p 22,80,443 [IP] - 특정 포트만 스캔

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -p 22,80,443 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:04 KST
Nmap scan report for 172.16.29.11
Host is up (0.00033s latency).

PORT    STATE  SERVICE
22/tcp  open   ssh
80/tcp  open   http
443/tcp closed https
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.20 seconds
```

nmap -sS [IP] - 스텔스 스캔(SYS Scan)

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -sS 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:07 KST
Nmap scan report for 172.16.29.11
Host is up (0.000088s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
3306/tcp open  mysql
8000/tcp open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds
```

→ 방화벽 우회 확률 증가. 실무 레드팀 기본.

nmap -f [IP] - Firewall 우회 (느리지만 은밀)

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -f 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:08 KST
Nmap scan report for 172.16.29.11
Host is up (0.00011s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
3306/tcp open  mysql
8000/tcp open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.33 seconds
```

→ 패킷을 쪼개 전송 (fragment)
단, 실무에서 드물게만 사용.

nmap 172.16.29.0/24 - 호스트 여러 개 스캔

```
┌──(zizon㉿zizon)-[~]
└─$ nmap 172.16.29.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:09 KST
Nmap scan report for 172.16.29.1
Host is up (0.00036s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT     STATE SERVICE
135/tcp  open  msrpc
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
3389/tcp open  ms-wbt-server
6000/tcp open  X11
MAC Address: 70:85:C2:72:C5:98 (ASRock Incorporation)

Nmap scan report for 172.16.29.11
Host is up (0.00011s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
3306/tcp open  mysql
8000/tcp open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap scan report for 172.16.29.20
Host is up (0.000094s latency).
Not shown: 991 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
80/tcp   open  http
81/tcp   open  hosts2-ns
111/tcp  open  rpcbind
139/tcp  open  netbios-ssn
445/tcp  open  microsoft-ds
2049/tcp open  nfs
8000/tcp open  http-alt
MAC Address: 00:0C:29:C4:0B:5D (VMware)

Nmap scan report for 172.16.29.33
Host is up (0.0044s latency).
Not shown: 993 closed tcp ports (reset)
PORT     STATE SERVICE
53/tcp   open  domain
```

nmap -T4 [IP]

```
┌──(zizon㉿zizon)-[~]
└─$ nmap -T4 172.16.29.11
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-24 20:12 KST
Nmap scan report for 172.16.29.11
Host is up (0.00012s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE
21/tcp   open  ftp
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
3306/tcp open  mysql
8000/tcp open  http-alt
MAC Address: 00:0C:29:ED:03:6B (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds
```

빠르게

Nmap의 핵심 중 하나.
웹, ssh, smb, ftp 등 서비스별 취약점 점검 스크립트 실행.

예)

nginx 코드 복사

```nginx
nmap --script http-enum 172.16.29.11
```

nginx 코드 복사

```nginx
nmap --script vuln 172.16.29.11
```

→ 자동 취약점 스캔 느낌.

14) 웹 서버 스캔에 유용한 옵션

nginx

코드 복사

```
nmap --script http-title 172.16.29.11
```

→ 웹 페이지 제목 확인

nginx

코드 복사

```
nmap --script http-headers 172.16.29.11
```

→ Apache/Nginx 버전 노출 여부 확인

nginx

코드 복사

```
nmap --script http-methods 172.16.29.11
```

→ PUT/DELETE 허용 여부 검사 (중요)

실무에서 nmap을 어떻게 쓰냐?

1) 침투테스트 / 레드팀

- 대상 서버 IP 범위 확인 (-sn)
- 서비스 식별 → 공격 벡터 결정 (-sV)
- 버전 기반 취약점 조사
- SSH? MySQL? FTP?
  → Hydra로 브루트포스 들어갈지 결정
- 웹이면 브프랑 연계
  → nmap으로 포트 확인 → burp로 인젝션/XSS 테스트

2) 블루팀

- 불필요하게 열린 포트 발견
- 잘못된 서비스 버전 확인
- 방화벽 정책 점검
- 서버가 공격받고 있는 흔적 탐지(스캔 로그)

3) 네트워크/시스템 운영

- 어떤 서버가 살아있는지 주기적 확인
- 포트 오픈 상태 점검
- 장비 교체 시 확인
- 서비스 죽었는지 체크 자동화